

PRACTICAL COURSE WORKBOOK

Ethical Hacking & Penetration Testing

Apply practical protection to accounts and systems

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a prioritised security review and action plan
SKILLS	Ethical hacking, Kali Linux, OWASP, CTF, Critical evaluation, Documentation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use Ethical hacking appropriately in a realistic, bounded task.
- Use Kali Linux appropriately in a realistic, bounded task.
- Use OWASP appropriately in a realistic, bounded task.
- Use CTF appropriately in a realistic, bounded task.

WHO THIS IS FOR

Defenders and developers learning lawful, defensive security practice.

BEFORE YOU BEGIN

- Basic networking and operating-system knowledge

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

Ethical Hacking & Penetration Testing is a structured, practical course covering Ethical hacking, Kali Linux, OWASP, CTF. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Ethical hacking: Security foundations	1. Assets with Ethical hacking 2. Threats with Kali Linux 3. Risk with OWASP
02 Kali Linux: Defensive controls	1. Identity with CTF 2. Network protection with Ethical hacking 3. Hardening with Kali Linux
03 OWASP: Assessment	1. Safe testing with OWASP 2. Evidence with CTF 3. Remediation with Ethical hacking
04 CTF: Operations	1. Monitoring with Kali Linux 2. Response with OWASP 3. Recovery with CTF

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Ethical Hacking & Penetration Testing project using Ethical hacking, Kali Linux, OWASP.

DELIVERABLE	A working Ethical Hacking & Penetration Testing artefact plus an evidence-based self-review.
TOOLS	A suitable Ethical hacking environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Ethical hacking.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

Use security techniques only on systems you own or are explicitly authorised to test.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Cybersecurity Framework 2.0

NIST - reviewed 2026-08-21
<https://www.nist.gov/cyberframework>

Web Security Testing Guide

OWASP Foundation - reviewed 2026-08-21
<https://owasp.org/www-project-web-security-testing-guide/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Ethical hacking scope.